

# B Lohini Priyanka

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## Basic Details

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**Date of Birth:** 27 Oct 1991

**Nationality:** Indian

**Ethnicity:** Tamil

**Gender:** Female

**Current Position:** PhD Scholar, IIIT Bangalore

## Research Vision

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Developing analytical frameworks for performance analysis and channel estimation in RIS-empowered, NOMA-enabled Integrated Sensing and Communication (ISAC) systems. Focused on scalable system design, mobility-aware performance evaluation, and intelligent RIS phase control for next-generation wireless networks.

## Education

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### IIIT Bangalore

2023–Present

PhD in Performance Analysis and Channel Estimation of RIS-empowered NOMA-enabled Integrated Sensing and Communication Networks (CGPA: 3.7/4.0)

Advisor: Prof. Priyanka Das

*Research Focus:* Machine Learning based solutions for channel estimation and RIS phase optimization in RIS-ISAC networks

### Thiagarajar College of Engineering

2013–2015

M.E in Wireless Technologies (CGPA: 9.54/10)

*Thesis:* Antenna Grouping based Feedback Compression and Channel Estimation for Massive MIMO-OFDM Systems

### Thiagarajar College of Engineering

2009–2013

B.E in Electronics and Communication Engineering (CGPA: 9.31/10)

*Project:* MIMO-OFDM channel estimation using Compressed Sensing

## Research Experience

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### Research Associate – IIIT Bangalore

Feb 2023–Dec 2023

Developed Sparse Bayesian Learning based Convolutional Deep Neural Network (SBL-DCNN) for channel estimation in mmWave MIMO-OTFS Systems.

Published the work in *IEEE WCNC*, 2025.

## Academic Experience

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### Assistant Professor – Veltech University, Chennai

Jun 2017–Nov 2018

## Publications

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- B. Lohini Priyanka, Priyanka Das, Amrita Mishra, ‘Performance Analysis of RIS-Assisted ISAC-NOMA Uplink System Under Imperfect SIC,’ -Accepted to be published in *IEEE WCNC 2026*, Kuala Lumpur, Malaysia. [PDF](#)

- B. Lohini Priyanka, Amrita Mishra, Priyanka Das, 'Convolutional Neural Network-Based Channel Estimation for mmWave MIMO-OTFS Systems,' - *IEEE WCNC 2025*, Milan, Italy. [PDF](#)
- B Lohini Priyanka K. K Rajeswari, S.J. Thiruvengadam, 'Compressed Sensing Based Channel Estimation in MIMO-OFDM System with Transmit Antenna Selection,' *International Journal of Applied Engineering Research*, Anna University, Chennai. Annexure II - 2014 (Sl.No.8565). [PDF](#)
- B Lohini Priyanka K. K Rajeswari, S.J. Thiruvengadam, 'MIMO-OFDM Channel Estimation using Distributed Compressed Sensing', *IEEE International Conference on Computational Intelligence and Computing Research- 2014*, Coimbatore, India. [PDF](#)
- B Lohini Priyanka K. K Rajeswari, S.J. Thiruvengadam, 'Sparse Channel Estimation using Compressed Sensing', *Fourth National Conference on Intelligent Techniques in Control Optimization and Signal Processing (INCOS)*, 2013., Tamil Nadu, India. [PDF](#)

## Technical Skills

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- AI/ML: Python (TensorFlow), Convolutional Neural Networks
- Tools: NS-2, Ansoft, Advanced Design System, MATLAB

## Certifications

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**Python for Machine Learning, Neural Networks and Deep Learning** (Department of Electrical Engineering, IIT Kanpur, December 2-22, 2023) [PDF](#)

**Joint Telematics Group (JTG)/ IEEE Information Theory Society Summer School in "Signal Processing, Telecommunication and Networking"** (IIT Madras, June 25-28, 2019)

**Key Enabling Technologies for 5G Wireless Networks** (GIAN, NIT Calicut, December 8 -13, 2017.) [PDF](#)

**Convex Optimization and Random Processes for Wireless Communication Systems** (TEQIP, Thiagarajar College of Engineering, March 05-07, 2015) [PDF](#)

**Computer Vision and Medical Image Processing: A Practical Approach using MATLAB** (TEQIP, Thiagarajar College of Engineering, April 25-26, 2014) [PDF](#)

**Advanced Signal Processing- Theory and Practice** (TIFAC-CORE in Wireless Technologies, Thiagarajar College of Engineering, August 10-13, 2011) [PDF](#)

**LabVIEW** (TIFAC-CORE in Wireless Technologies, Thiagarajar College of Engineering, June 7-10, 2011) [PDF](#)